

Capstone 2 Presentation Outline

Capstone 2 Presentation: Diabetic Readmission Risk

Slide 1: Title & Team

- Project Title: Predicting 30-Day Hospital Readmission in Diabetic Patients
- Group Name: Diabetes Insights Team
- Members: Raman; Abhijeet; Vaibhav Singh; Ashish Singh Naruka; Ayush Shukla

Slide 2: Business Problem

- Question: What clinical factors predict 30-day readmission?
- Impact: CMS penalties and \$26B annual cost in the US.

Slide 3: Data ETL & Quality

- Source: UCI ML Repository (101,766 rows).
- Process: ? replacement, duplicate removal (30k rows), age mid-point conversion.

Slide 4: Key EDA Findings

- Admission Type: Emergency admissions have highest risk.
- Polypharmacy: Patients on 20+ meds are high risk.
- Insulin: Dose changes (Up/Down) indicate instability.

Slide 5: Statistical Analysis

- Chi-Square: Validated Admission Type as significant.
- LogReg: Multivariable model outperforms single features.
- ANOVA: Emergency stays are significantly longer.

Slide 6: Tableau Dashboard

- Interactive filters for Clinicians.
- Risk Heatmap (Age vs Admission Type).

Slide 7: Business Recommendations

1. Emergency risk scoring.
2. Medication reconciliation.
3. Insulin stability window.

Slide 8: Q&A

- Thank you!